26900049 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Skill Enhancement Course (SEC) - MG5SECBIF300 - INTRODUCTION TO R PROGRAMMING

(2024 ADMISSION ONWARDS)

Duration: 1 hour **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **5** questions

Each question carries **2** marks

1. Differentiate between a matrix and a data frame. [U] / [CO3]
2. Explain the role of next. [U] / [CO3]
3. Who developed the R programming language? [K] / [CO1]
4. What is a character constant in R? [K] / [CO1]
5. Analyse the role of R objects in storing and retrieving data from files. [An] / [CO5]
6. Evaluate the usefulness of histograms generated using R for analysing data distribution. [A] / [CO6]
7. Write the syntax of an if statement. [K] / [CO1]

[2x5 = 10]

Part B

Short Essay

Answer any **3** questions

Each question carries **5** marks

8. Write an R program to create a factor representing student grades. [A] / [CO3]

9. Explain different methods of accessing list elements. [U] / [CO3]

10. Explain readline() and scan() with suitable examples. [U] / [CO2]

11. Analyse the role of the R plot() function in data visualization. [An] / [CO4]

12. Analyse the methods available in R for reading CSV, Excel and other file formats. [An] / [CO5]

[5x3 = 15]

Part C

Long Essay

Answer any **1** questions

Each question carries **10** marks

13. Write an R program to create a student data frame and perform various manipulation operations. [A] / [CO3]

14. Explain the different types of operators available in R with suitable examples. [U] / [CO1]

15. Evaluate the effectiveness of pie charts, scatter plots and line charts in communicating information from a dataset. [E] / [CO4]

[10x1 = 10]